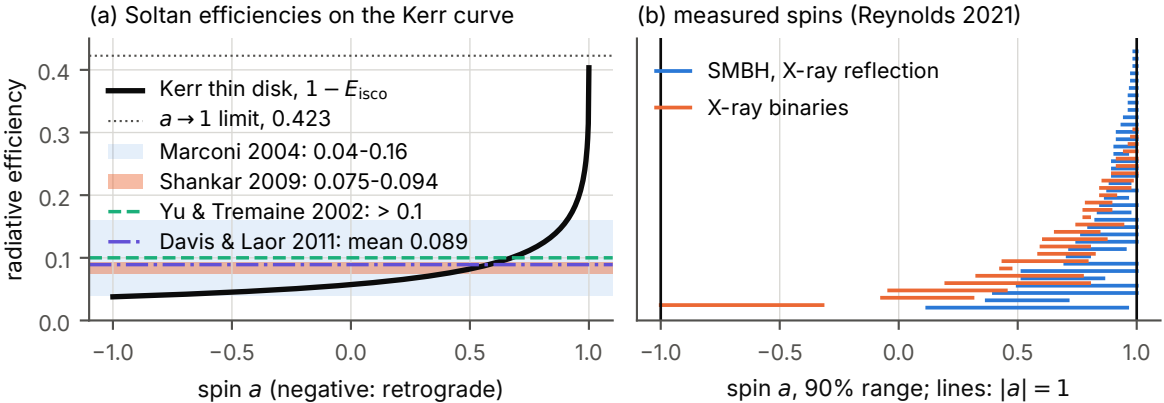
(a) Soltan efficiencies on the Kerr curve
radiative efficiency
0.0
0.1
0.2
0.3
0.4
Kerr thin disk, $1 - E_{\rm isco}$
$a \to 1$ limit, 0.423
Marconi 2004: 0.04-0.16
Shankar 2009: 0.075-0.094
Yu & Tremaine 2002: > 0.1
Davis & Laor 2011: mean 0.089
−1.0
−0.5
0.0
0.5
1.0
spin a (negative: retrograde)
(b) measured spins (Reynolds 2021)
SMBH, X-ray reflection
X-ray binaries
−1.0
−0.5
0.0
0.5
1.0
spin a, 90% range; lines: $|a| = 1$